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Item Learnability and Subsequent Memory Prediction

1.1 INTRODUCTION

When studying for a test in a language class, students might find that they easily recall some words but not others. What is it that makes a particular word or image (or more generally,

item) more likely to be remembered than another? Behavioral studies of memory have uncovered several mechanisms by which an item might be more strongly encoded such as the spacing between multiple study sessions (Ruch, 1928), the order in which items are studied (Murdock, 1962) or the task used to encode the items (Craik and Tulving, 1975). In addition to these variables, items themselves (both words and images) appear to have intrinsic, measurable differences in their average *memorability* - the general rate at which the average person will remember them (Rubin, 1985; Rubin and Friendly, 1986; Baldwin and Runkle, 1967; Isola et al., 2011). At least some of these item-specific effects can be accounted for by features like the distinctiveness of a face (Light et al., 1979; Going and Read, 1974) or the frequency (DeLosh and McDaniel, 1996) or semantic centrality (Xie et al., 2020) of a word. However, there remains substantial variability in memory performance even after accounting for these factors (Kahana et al., 2018; Bainbridge, 2019). The goal of the present report is to understand this additional variability.

The advent of cognitive neuroscience has opened up the possibility of measuring neural signals during the act of memory encoding or learning that may account for this additional variation in what gets remembered. The key idea is that neural activation patterns recorded during encoding may reflect the success or durability of new learning. The dominant approach to finding these signals has been the “subsequent memory paradigm” (Paller and Wagner, 2002), where neural signals collected while a subject tries to encode several items are compared based on later memory performance (e.g., comparing items that were later remembered to those that were later forgotten). While such findings were first reported using ERPs (Sanquist et al., 1980; Paller et al., 1987), these were soon followed by fMRI studies, which had the ability to measure differences in neural activity with high spatial

precision in healthy adults (Wagner, 1998; Davachi et al., 2003; Ranganath et al., 2003). Across these studies a coherent set of brain regions have been identified as being involved in human memory formation including the hippocampus and medial temporal lobe (Kim, 2011), which have long been associated with memory formation on the basis of animal and lesion studies (Scoville and Milner, 1957; Mishkin, 1978). In recent years, the field has moved beyond univariate activity, showing that the similarity of multivariate activity across presentations of the same item or other studied items is also related to memory recall and recognition (Rissman and Wagner, 2012; Xue, 2018) and suggesting particular computations that might be involved in memory formation.

Overall, the success of these studies would suggest that fMRI signals may account for additional variability in encoding processes beyond those identified above. Indeed, many researchers hope to interpret these types of subsequent memory effects causally, claiming that they measure processes that make an item more or less likely to be remembered. For instance, at least some researchers claim that certain subsequent memory analyses identify neural signals that index “moment-to-moment fluctuations of the quality of memory encoding” (Fukuda and Woodman, 2015) or activity “that supports successful encoding... [and] that leads to encoding failure” (Kim, 2011). However, recent work has suggested that the *memorability* of a particular item may drive a significant amount of brain activity during memory encoding (Bainbridge et al., 2017) and retrieval (Bainbridge and Rissman, 2018). The concept that item memorability might itself modulate neural processes poses a significant challenge to the causal interpretation of subsequent memory effects because it leads to a classic omitted variable bias: features that make an item memorable may lead to increased activity in brain regions that are unrelated to the memory encoding mechanisms

per se. For example, easier-to-remember items might be more emotional or vivid and this emotional content might drive neural activity that is unrelated to the specific operations involved in memory formation. This activity would then be *correlated* with and predictive of subsequent memory but lack any causal link to the strength of encoding or the formation of an “engram.”

In general, understanding and properly interpreting subsequent memory effects requires decomposing these signals into those that are driven by external factors, such as those uncovered by psychological studies mentioned above, including memorability, and those that can only be explained by endogenous factors occurring within the brain of individuals (Weidemann and Kahana, 2019). In particular, the effects of endogenous shifts in attention or other variation in the quality or fidelity of a representation during encoding (perhaps due to variation in, e.g., personal semantic associations (Chadwick et al., 2016)) are often thought to contribute to subsequent memory effects, especially multivariate ones (Xue, 2018). Weidemann and Kahana (2019) recently argued that subsequent memory effects found in oscillatory activity (as measured by scalp EEG) are composed of a mix of both endogenous and exogenous factors. Deconfounding these effects in fMRI with known effects of serial position or item memorability is key to interpreting several classic results concerning the contribution of different brain regions to the cognitive neuroscience of memory.

Deconfounding these various contributing factors has the potential to affect fields outside of cognitive neuroscience as well. One of the original motivations (Sanquist et al., 1980) for searching for subsequent memory effects was that traditional cognitive psychology measurements like response times failed to cleanly capture the depth of encoding of an item (Craik and Tulving, 1975). Measures of subsequent memory have the potential

to fulfill this role and have already been used to make claims about cognitive mechanisms of learning such as the spacing effect (Feng et al., 2019). However, for a neural measure to provide a useful measurement of encoding processes, it would have to be able to predict individual variation in encoding after accounting for population-level effects on recall such as item and position. If we were to find such a signal, it could prove very useful for future cognitive modeling of learning and memory, particularly with modern joint modeling techniques (Turner et al., 2013, 2019). In addition, having independent measurements of learning are of great interest not only in basic research but also in more applied fields as they have the potential to inform educational interventions and automated tutoring systems (Atkinson, 1972; Corbett and Anderson, 1995).

1.1.1 HOW TO TELL IF A BRAIN REGION CONTRIBUTES TO MEMORY FORMATION

Before describing our specific studies, it is helpful to elaborate on the standards of evidence that would allow one to conclude that a particular brain region is involved in endogenous contributions to memory formation.

First, consider the standard subsequent memory paradigm. In this approach neural activity is recorded from one or several regions of the brain while a subject is learning. Later, the subject is given a memory test. The neural recordings for items that were later forgotten are then compared to those that were later remembered. Significant differences in activation between these regions are often taken as an index of the neural signatures of successful memory formation.

As mentioned earlier however, this approach has several limitations. To the degree that brain activity is driven by the content of individual items, then some of this activity might

reflect item specific features, such as vividness, uniqueness and emotional intensity, which is generally expected to be higher for more memorable stimuli (Bainbridge, 2019). This raises an important but often overlooked point in the literature on subsequent memory. A region should be endorsed as being involved in the endogenous processes related to memory encoding to the degree that it predicts memory performance over and above that which might be attributed to the item itself. We would therefore like to be able to control for the memorability of an item in our analysis and identify brain signals that are able to predict memory, given two items that are otherwise equal in their expected probability of remembering, based on estimates from a larger population.

Having laid out some initial criteria to handle confounding, there is still a question of which of several regions of the brain should be tested. Even among regions shown to predict memory in standard analyses (Kim, 2011), we do not know which ones are specifically related to endogenous processes and which ones are largely explained by memorability. We also expect that these signals may be quite small because some of the variance is already accounted for by memorability. In order to address these issues, we use a predictive modeling approach. Specifically, we attempt to predict if a subject will correctly recall an item using signals from several brain regions rather than the standard explanatory approach where signals in individual brain regions are predicted based on whether they were later recalled (Yarkoni and Westfall, 2017). By training a predictive model and evaluating its performance in held-out participants, we are able to select among many features without being concerned about reduced power from multiple comparisons. Across-subject prediction increases power to find small effects relative to within subject prediction (Wang et al., 2020), particularly when they are fairly consistent across subjects as we might expect for a signal of

an attentive state. Furthermore successful prediction in a held-out subject suggests that a feature is robust and useful for predicting memory formation.

To sum up, the bar that we propose to adopt is one where a region is considered as distinctly contributing to endogenous memory operations when:

- The feature can be used to predict the behavior of a held-out person
- The feature is useful for prediction above and beyond a model that accounts for the overall difficulty of learning in the general population (i.e., learnability or memorability).

The first condition ensures reliable memory prediction while second condition addresses omitted variable concerns that allow us to better interpret the signals. This might seem a very high bar to set, however, given that the basic phenomena of neural patterns facilitating subsequent memory has been largely established in the literature, it seems time for an assessment of the unique predictive value of such features. In addition, [Weidemann and Kahana \(2019\)](#) recently successfully applied a similar approach to identifying EEG signals, suggesting that this bar is indeed reachable with modern methods.

1.1.2 OVERVIEW OF THE PRESENT WORK

The present study was designed to explore these issues in a novel and somewhat unique fMRI study of memory in healthy adults. We designed and collected a new data set for several reasons. First, while there is a large literature on subsequent memory effects many of the classic studies were conducted several years ago and are thus relatively small. This makes it difficult to conduct analyses using a predictive approach which requires holding out

some amount of data from model estimation. In addition, modern fMRI methods have progressed significantly in the past several years, increasing spatial and temporal resolution and making new data sets potentially more powerful.

Second, one motivation for this project was evaluating the potential for subsequent memory effects to make predictions about memory in contexts that are more typical of everyday classroom learning. In the psychology literature on memory, a Lithuanian-English vocabulary learning task has attracted considerable study partly due to the presence of large scale norming studies which have calibrated the overall difficulty of various items (Grimaldi et al., 2010; Nelson et al., 2016). Unlike many subsequent memory studies, we designed a study that involved several repeated presentations of the items. This offered several benefits including increasing the overall accuracy after a long delay, ensuring that there is significant variability to explain; also, it allowed us to measure multivariate fMRI signals thought to predict memory that involve comparing repeated presentations of the same item (Xue et al., 2010; Xue, 2018). Finally, we adopted one rather unusual step in our design by presenting all the words to subjects in the same exact order. Most studies of memory randomize item presentation to attempt to control for item order effects. While item order and context is confounded in our study, this allows us to more precisely quantify the effects of item memorability, presentation order, and their interaction on subsequent recall, at the expense of generalizability. In addition, several recent studies have shown that correlated activity across subjects is predictive of memory in many settings (Hasson et al., 2008; Koch et al., 2020; Chen et al., 2017). However, these have previously only been tried on engaging video stimuli that are likely to align the time course of the fMRI BOLD signal and deal with autoregressive noise. By using the same presentation order for every subject, we

are able to treat our stimuli as essentially video-like which allows us to precisely align time courses and test this feature’s ability to predict memory in our task as well.

To give a high level overview, we scanned subjects while they engaged in a Lithuanian-English second-language learning task where pair of words consisting of an English word and its Lithuanian translation were presented one at a time. In order to identify signals that *specifically* reflect endogenous learning or memory processes, we first estimate a baseline predictive psychometric model, that only includes information about the item being studied and the subject studying it. We then estimate several models that include the prediction from this baseline model and various fMRI signals thought to relate to subsequent memory (Xue, 2018). We use leave-one-subject-out cross validation to evaluate these models on their ability to rank words based on their probability of recall, using the area under the receiver operating characteristic curve (AUC) as a metric. AUC is ideal for our purposes because alternative metrics, such as log likelihood, can be improved even if the model only reflects a subject’s overall ability rather than endogenous fluctuations in attention that is specific to an item. Therefore, a significant increase in AUC in the models that include fMRI signals suggests that a particular feature passes our test laid out above.

Perhaps the most important question is what neural features or signals are most useful for the prediction task as we have defined it. Here there are not necessarily clear answers. Several features of the fMRI BOLD signal and regions of interest (ROIs) have been implicated in memory operations in past work. However, due to differences in stimuli and tasks design it is not always clear when these patterns should apply to a new study. As a result we attempted a systematic literature review drawing from several possible neural features and ROIs that might apply to our study and can be measured at the time of encoding. The

following sections describe at a high level some of the features we considered.

NEURAL SIGNALS

While early fMRI studies on subsequent memory effects focused on demonstrating differences in univariate activation (Kim, 2011; Paller and Wagner, 2002), subsequent work expanded the types of signals studied to include multivariate patterns (Rissman and Wagner, 2012; Xue, 2018). In the following, we define the three types of multivariate signals we will test with our approach in addition to univariate activity. These signals' ability to predict memory has been extensively documented in the past decade (Xue, 2018) and involve comparing the spatial pattern of voxel activation across time points or across people (commonly known as pattern similarity (Kriegeskorte, 2008; Edelman et al., 1998)).

Univariate Activation Univariate activity is by far the most studied subsequent memory effect and, as described above, several canonical regions appear to correlate with later memory (Kim, 2011; Paller and Wagner, 2002). Activity in frontal regions (Kirchhoff et al., 2000) is often thought to be related to processing and interpreting sensory input to make it suitable for encoding whereas activity in the medial temporal lobe is related to the act of binding inputs with contexts or other items for future retrieval (Brown and Aggleton, 2001; Davachi, 2006). Finally, activity in frontoparietal regions is thought to reflect attention and selection of items for storage (Cabeza et al., 2008). The appearance of greater activity in all of these regions is typically interpreted as signalling to signal greater depth of processing, increased alertness and stronger binding which is why they are all thought to predict greater encoding strength. There also exist regions whose activity decreases during successful encoding, typically associated

with the default mode network (Otten and Rugg, 2001; Park and Rugg, 2008). This change in activity is typically interpreted as a decrease in mind-wandering and focus on internal thoughts although it's meaning is still up for debate. Another reason for a relationship between univariate activity and later memory in our task is repetition suppression where regions involved in processing a stimulus show decreased activity on repeated presentations (Grill-Spector and Malach, 2001). While this has been shown to predict implicit memory, evidence for its relationship with explicit memory tasks like ours is more mixed (Ward et al., 2013). That being said, it is possible that we will see a relationship in our task.

Item Pattern Similarity (IPS) The pattern similarity of an item with itself across repeated presentations has been shown to be correlated with memory in a number of studies, using both visual and verbal stimuli in recognition memory, associative fear learning, cued recall and free recall tasks (Xue et al., 2010, 2013; Ward et al., 2013; Visser et al., 2013; Bruett et al., 2020). Measuring item-to-item pattern similarity requires multiple presentations of the same item. Our study involves five repeated presentations so it presents an ideal opportunity to study these effects. While the exact role that item pattern similarity plays in memory encoding is still unclear, the positive association between this feature and later memory in several ROIs (Xue et al., 2010) has been taken as evidence against the *encoding variability hypothesis* that repeated presentations improve memory by associating an item to several diverse contexts (Martin, 1968). In contrast, it has been suggested that an item's similarity to itself or other category members (Kuhl et al., 2012), suggests higher representational fidelity which is what promotes future memory (Xue, 2018). However, the exact role that

fidelity and diversity play in memory encoding is still under active debate (Sievers et al., 2019; Kılıç et al., 2017)

Global Pattern Similarity (GPS) In many regions, the similarity of a presentation of an item with all other studied items or all remembered items, has been shown to positively correlate with recognition memory with both verbal and visual stimuli (Bainbridge et al., 2017; Davis et al., 2014; LaRocque et al., 2013; Xiao et al., 2016). Global pattern similarity as a predictor of memory has often been theoretically motivated by models of recognition memory (Norman and O'Reilly, 2003; Gillund and Shiffrin, 1984; Nosofsky, 1988; Hintzman, 1988). While models differ in the details, in general, activation of an item in memory depends on how similar it is to a retrieval probe. Therefore, the degree to which an item is similar to many other studied items makes it more likely for lots of items to be activated when probed and therefore, for the item to seem familiar on a recognition test. However, there is good reason to think that it might be predictive of other kinds of memory as well. For instance, both hippocampal pattern separation and similarity across items is thought to increase the probability of retrieval in various memory tasks (Brunec et al., 2020). LaRocque et al. (2013) found that global pattern similarity in hippocampus was negatively correlated with recognition memory, which may be true across other tasks like cued and free recall. In addition, Kuhl et al. (2012) found that within-category similarity at encoding predicted later memory for category in a cued recall task. Because the categories in their task (faces and places) are broadly associated with processing in particular brain regions (FFA and PPA), this predictive power could just be indexing proper processing of the stimuli. To the extent that the English and

Lithuanian word pairs need to be processed both semantically and phonologically, we can consider all items in our study to be a part of a similar “category” or type of stimulus that requires similar processing.

Inter-subject Pattern Correlation (ISPC) Several recent studies have explored the shared variance in neural signals while watching engaging stimuli like naturalistic videos (Chen et al., 2017). The across-subject temporal correlation of the same voxel (inter-subject correlation, Hasson et al. (2008)) and the spatial correlation of the average pattern in an ROI (intersubject pattern correlation Koch et al. (2020)) while both subjects watch the same video are two signals that have been shown to correlate with later recollection of those videos. While our task did not use video stimuli, subjects observed items in the task in the exact same order with the exact same timing. This suggests that it may be reasonable to treat our data in a similar way to data using video stimuli as responses to the stimuli should line up temporally.

REGIONS OF INTEREST

In order to test our ability to predict memory, we compare classifiers that each contain features from one of the feature sets described above (univariate activity, item pattern similarity, global pattern similarity, inter-subject pattern correlation). All features involve * aggregating over a set of voxels and the same feature in several different regions may predict recall, possibly for different reasons (Xue, 2018). Ideally we aim to strike a balance between including all potentially relevant signals and not including too many unrelated features. We

*While it is possible that single voxel univariate activity is a good feature for predicting memory, it is highly unlikely that single voxels will perfectly align across all of our subjects. Therefore, we aggregate over a set of voxels even for univariate activity

therefore test classifiers using the following sets of regions described below.

Targeted ROIs We first test a set of cortical ROIs derived from the existing cognitive neuroscience literature. We defined a targeted set of nineteen ROIs on the basis of prior expectations of these regions’ potential engagement in a learning task involving verbal materials. We selected left and right perirhinal cortex (PRC) and anterior temporal lobe (ATL) ROIs due to putative roles for these regions in processing conceptual information (Martin et al., 2018). In addition, we selected regions highly likely to be involved in memory formation (Kim, 2011) and in processing concrete nouns (a category which includes all of the word pairs in our stimulus set) (Chen et al., 2019). These ROIs included: two separate areas in left anterior insula, left and right lateral occipital cortex, left and right dorsal parietal, left and right lateral parietal, left and right medial parietal (precuneus), left and right retrosplenial cortex, an upper and a lower portion of left ventrolateral prefrontal cortex (VLPFC), and visual word form area (VWFA; left hemisphere only).

Hippocampus In a separate set of classifiers, we test features in a hippocampus ROI defined based on each participant’s individual anatomical scan. We test this separately both because of the central role the hippocampus is thought to play in memory formation (Scoville and Milner, 1957; Marr, 1971; Norman and O’Reilly, 2003) and because of the difficulty in recording fMRI BOLD signal from the hippocampus when using standard imaging sequences for targeting the whole brain as we do in this study (Litman and Davachi, 2008). The separate set means that the smaller signal-to-noise ratio in hippocampus BOLD will not prevent its inclusion in the penalized regression models we use for prediction (described in more detail below). We

test classifiers that include the average activity, pattern similarity and global pattern similarity in each individual’s whole hippocampus (*hipp-whole*) and also hippocampal subparts (left and right, posterior, medial and anterior, *hipp-sub*). In addition, we also select the voxels that are included in all participant’s anatomically-defined hippocampus which allows us to test a classifier based on individual hippocampal voxel activity *hipp-vox*). Finally, we can define a whole hippocampus (*hipp-whole*) inter-subject pattern correlation feature based on these overlapping voxels.

All Cortical ROIs In addition to these two sets of theoretically motivated ROIs, we also include a set of 100 cortical ROIs defined based on resting-state functional connectivity (Schaefer et al., 2018). This allow us to be sensitive to any possible signals that might exist outside of the previous sets of ROIs.

Whole Brain Finally, one challenge in across-subject prediction is that voxels may not be perfectly aligned (Wang et al., 2020; Haxby et al., 2020). Therefore, in order to be less sensitive to these differences, we follow a strategy from Rissman et al. (2010) where we treat the average activity in an ROI as a “voxel” for the purpose of computing these features. We then compute the similarity of activity across ROIs in all of cortex for use in our multivariate features. The ROIs we use here come from the 100 cortical ROIs from Schaefer et al. (2018), also described in the previous set.

1.2 METHODS

1.2.1 PARTICIPANTS

Sixty-nine participants were recruited from using electronic advertisements hosted by New York University and accessible to the broader community. We obtained written informed consent prior to conducting the study, in compliance with the New York University Institutional Review Board. All participants self reported to be between 18-35 years old, had normal or corrected-to-normal vision, spoke English fluently and did not speak Lithuanian or a related language.

Eleven participants' data were excluded from analysis based on MRI data issues: two participants' MRI sessions were conducted with incorrect scanning parameters; six participants were excluded based on excessive motion and other image quality issues; three participants requested early termination of the experiment due to discomfort in the scanner (and did not complete the recall test). We also excluded one of the participants who recorded 100% correct in the recall test since their held-out AUC would be undefined, resulting in a final data set of fifty-seven participants

1.2.2 BEHAVIORAL TASK

Based on a normed set of Lithuanian-English words, we selected 45 translation pairs with a range of difficulties (Grimaldi et al., 2010). Participants first completed a study phase inside the fMRI scanner where they saw the translation pairs presented one at a time for 4 seconds each with a variable duration inter-trial interval (randomly chosen between 4 and 16 seconds). During this inter-trial interval, in order to prevent rehearsal, participants made

judgments about whether each of a series of numbers were odd or even.

Words were presented on a computer screen with the Lithuanian word at the top of the screen and the English translation underneath. Each word pair was presented five times and no pair was presented for the n th repetition until all words had $n - 1$ presentations. Importantly, and in contrast to many psychology studies on the subsequent memory effect, all participants see the same sequence of study items. At the expense of generalizability, this allowed us to precisely quantify the average recall performance for an word pair in a specific context and accounting for any interactions between items. The order of the word pairs was selected as follows: the 45 words were grouped into groups of five words each. On each repetition the order of the five words were shuffled but all five words appeared before the next group of five were presented. Immediately following the study session participants completed a second section outside the scanner that involved making judgments of learning (JOLs, (Nelson and Dunlosky, 1991)): for each pair participants were presented with the Lithuanian and English word and used the computer mouse to indicate on a scale of 0-100 how likely they were to remember the association in three days. Participants had up to twelve seconds to respond and the response was coded as missing if this deadline wasn't met (in practice all participants gave all JOL ratings within the allotted time).

Participants were then asked to complete a recall test approximately 72 after the initial scan session. The first 44 of the subjects completed this in the lab and the final 13 subjects completed the second session at home, via an online version of the task to ease the task of scheduling sessions with a 72 hour delay between. Participants saw a Lithuanian word presented on the screen and had to type the associated English word. A trial was coded as correct if participants typed the correct English word (allowing for typographic errors) and

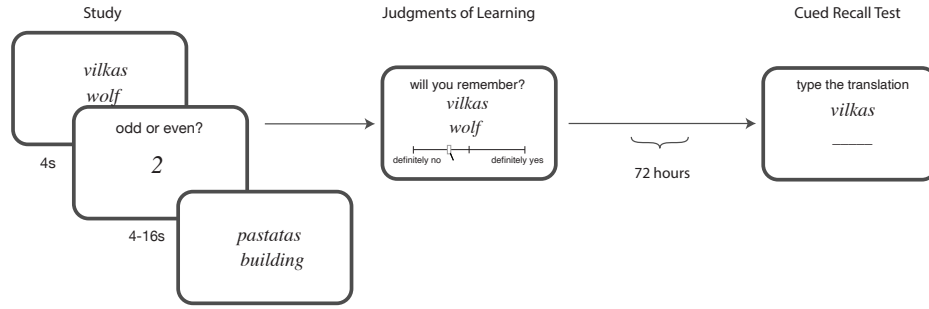


Figure 1.1: Experimental Design. On Day 1, in the fMRI scanner, subjects first completed a paired associates learning task in which an English word and a Lithuanian word were simultaneously presented on the screen. In between trials, subjects completed an odd-even judgment about a series of numbers in order to prevent rehearsal. After the learning session, subjects made a judgment of learning about each of the paired associates outside of the scanner. On Day 2, 72 hours after Day 1, subjects completed a cued recall test in which they were presented with a Lithuanian word and typed in the corresponding English word.

all other responses were incorrect. Like the judgments of learning, this task was completed outside the scanner.

1.2.3 FMRI METHODS

MRI DATA ACQUISITION

MRI data were acquired on a 3 Tesla Siemens Prisma scanner. Anatomical images were collected using a T₁-weighted MPRAGE high-resolution sequence (0.8mm isotropic voxels). Five runs of whole brain BOLD functional data were acquired using an EPI sequence (2.5mm isotropic voxels; TR=1s; TE=35ms; Multiband factor=4; Phase Encoding: Anterior-Posterior). An additional pair of non-accelerated EPI scans with opposing forward (anterior-posterior) and reverse (posterior-anterior) phase encoding relative to the primary functional data were collected for distortion correction during preprocessing (2.5mm isotropic voxels; TR=4.496; TE=45.6). The slice position and orientation for these scans matched those of the BOLD data collected during the behavioral task.

PREPROCESSING

Results included in this manuscript come from preprocessing performed using fMRIPrep 1.2.6-1 (Esteban et al., 2019), which is based on Nipype 1.1.7 (Gorgolewski et al., 2011).

ANATOMICAL DATA PREPROCESSING The T1w anatomical scans were corrected for intensity non-uniformity (INU) using N4BiasFieldCorrection (ANTs 2.2.0, Tustison et al., 2010). A T1w-reference map was computed (after INU-correction) using `mri_robust_template` (FreeSurfer 6.0.1, Reuter et al., 2010). The T1w-reference was then skull-stripped using `antsBrainExtraction.sh` (ANTs 2.2.0), using OASIS as target template. Brain surfaces were reconstructed using `recon-all` (FreeSurfer 6.0.1, Dale et al., 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (Klein et al., 2017). Spatial normalization to the ICBM 152 Nonlinear Asymmetrical template version 2009c (Fonov et al., 2011) was performed through nonlinear registration with `antsRegistration` (ANTs 2.2.0, Avants et al., 2011), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using `fast` (FSL 5.0.9 Zhang et al., 2001).

FUNCTIONAL DATA PREPROCESSING For each of the five BOLD runs collected per subject (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. A deformation field to correct for susceptibility distortions was estimated

based on two echo-planar imaging (EPI) references with opposing phase-encoding directions, using 3dQwarp (AFNI 20160207, [Cox, 1997](#)). Based on the estimated susceptibility distortion, an unwarped BOLD reference was calculated for a more accurate co-registration with the anatomical reference.

The BOLD reference was then co-registered to the T1w reference using `bbregister` (FreeSurfer) which implements boundary-based registration ([Greve and Fischl, 2009](#)). Co-registration was configured with nine degrees of freedom to account for distortions remaining in the BOLD reference. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using `mcflirt` (FSL 5.0.9, [Jenkinson et al., 2002](#)). The BOLD time-series were resampled onto their original, native space by applying a single, composite transform to correct for head-motion and susceptibility distortions. These resampled BOLD time-series will be referred to as preprocessed BOLD in original space, or just preprocessed BOLD. The BOLD time-series were resampled to MNI152NLin2009cAsym standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space.

Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by [Power et al., 2014](#)). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor, [Behzadi et al., 2007](#)). Principal components are estimated after high-pass filtering the preprocessed BOLD

time-series (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). Six tCompCor components are then calculated from the top 5% variable voxels within a mask covering the subcortical regions. This subcortical mask is obtained by heavily eroding the brain mask, which ensures it does not include cortical GM regions. For aCompCor, six components are calculated within the intersection of the aforementioned mask and the union of CSF and WM masks calculated in T1w space, after their projection to the native space of each functional run (using the inverse BOLD-to-T1w transformation). The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction, and co-registrations to anatomical and template spaces). Gridded (volumetric) resamplings were performed using `antsApplyTransforms` (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Non-gridded (surface) resamplings were performed using `mri_vol2surf` (FreeSurfer). Preprocessed BOLD data were smoothed to 6mm FWHM using AFNI `3dBlurToFWHM`[†].

REGIONS OF INTEREST (ROIs)

Gray matter and hippocampus anatomical masks Analysis of the functional data was restricted to voxels encompassed by a binary mask computed from the probabilistic gray matter estimates generated during the FreeSurfer anatomical preprocessing combined with anatomically defined voxels in individual hippocampi. For each

[†]https://afni.nimh.nih.gov/pub/dist/doc/program_help/3dBlurToFWHM.html

participant, the probabilistic gray matter mask was binarized with threshold of 0.2 and intersected with bilateral hippocampal voxels as estimated using FSL FLIRT implemented in Nipype (Jenkinson and Smith, 2001; Jenkinson et al., 2002). The combined gray matter/hippocampus mask was resliced to functional resolution, smoothed with an 8mm FWHM kernel to be liberal in voxel inclusion (we opted to include some potentially non-gray matter voxels that could be ignored in downstream analyses rather than being overly strict and excluding relevant signal) and then re-binarized with a threshold of 0.2.

Whole brain cortical ROIs For whole brain cortical analysis, voxels were aggregated into ROIs provided by (Schaefer et al., 2018). To strike a balance between regional specificity and total number of cortical ROIs we used the intersection of the gray matter mask defined during preprocessing and the 100 parcel, 7 network atlas from Schaefer et al. (2018)[‡] after reslicing to $2.5mm^3$.

Targeted ROIs The set of targeted regions listed above were defined from a variety of sources. The PRC regions were defined using the anterior portion ($MNI y > -21$) of the PRC ROIs provided by (Ritchey et al., 2015) as downloaded from Neurovault[§] which overlapped with the PRC center coordinates reported by Martin et al. (2018). The ATL ROIs were defined as the Schaefer et al. (2018) ROIs encompassing the peak voxels reported by Martin et al. (2018) (Schaefer et al. (2018) 600 parcellation ROIs 191 and 492).

The remaining targeted ROIs were defined by identifying voxel groupings from

[‡]https://github.com/ThomasYeoLab/CBIG/tree/master/stable_projects/brain_parcellation/Schaefer2018_LocalGlobal

[§]<https://neurovault.org/collections/3731/>; <https://identifiers.org/neurovault.collection:3731>

Schaefer et al. (2018) which encompassed other cortical regions expected to be involved in memory formation and/or processing of verbal materials.

SINGLE TRIAL ACTIVATION ESTIMATES Individual voxel BOLD activation on each study trial was estimated with a general linear model (GLM) as implemented in SPM12 (Friston et al., 1994)[‡] using the *least squares-single* approach described in Mumford et al. (2012). For each study trial, a voxel-wise regression was carried out with one predictor containing a four second boxcar covering the timepoints corresponding to the to-be-estimated trial, another predictor containing four second boxcars covering the timepoints for all of the remaining trials, and fourteen additional predictors containing a variety of other potential confounds and nuisance signals as estimated during preprocessing (Power et al., 2014). Prior to estimation, the vectors of single-trial and all-other trials were convolved with the canonical SPM double gamma hemodynamic response function (HRF) along with its' temporal and dispersion derivatives (Lindquist et al., 2009). We then estimated regression coefficients for each voxel using ordinary least squares and the procedure was repeated for each study trial for each participant. The resulting coefficients for the single-trial predictors were taken as summaries for the activation for each voxel on each trial.

PATTERN SIMILARITY The three multivariate signals involve estimating distances between spatial patterns of activity across trials within a subject or across subjects. These patterns are derived from the single trial estimates for each voxel in a particular ROI. While several distance metrics have been proposed (Bobadilla-Suarez et al., 2019; Walther et al., 2016), in order to remain faithful to the proposed features, we use the Pearson's r , which is

[‡]<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>

the metric used by nearly every paper in the subsequent memory literature (e.g. [LaRocque et al., 2013](#); [Xue et al., 2010](#); [Davis et al., 2014](#)). In addition, for inclusion in the model, all of the features are Fisher- z transformed ([Fisher, 1915](#)) which is often used when aggregating correlations since it makes the sampling distribution approximately normal, which is slightly better for use in the regularized regression models we describe in the following section.

PREDICTIVE MODELING METHODS

After initial fMRI preprocessing, additional steps to create the fMRI features and reshape data were conducted using custom code in python 3.7.7 using the packages pandas ([McKinney, 2010](#)) and numpy ([van der Walt et al., 2011](#)) and R 4.0.2 using the package collections tidyverse ([Wickham et al., 2019](#)) and tidymodels ([Kuhn and Wickham, 2020](#)).

Baseline Behavioral Model In order to control for item memorability, we first estimate a baseline model that accounts for word and presentation-order effects. To do so, we use a one-parameter item-response theory (IRT) model:

$$P(r_{ws} = 1|s, w) = \text{logit}^{-1}(\theta_s + b_w)$$

$$\theta_s \sim N(0, \sigma_\theta)$$

where w indexes Lithuanian-English word pairs, s indexes subjects, r_{ws} is the response (correct or incorrect recall) of each subject to each word pair and b_w and θ_s represent the latent word memorability and subject ability parameters respectively. We estimate this model using maximum likelihood, as implemented in the R package lme4

(Bates et al., 2015). In summary, this model estimates one parameter for each subject which captures their overall ability, and one parameter per word-pair which captures its overall difficulty. As a baseline, the combination of these two factors already accounts for a substantial amount of the structure of the data that is measurable without looking at the neural features.

Full Model We use logistic regression as a model for predicting recall from the various proposed fMRI signals. Because our goal was to find features that improve predictions beyond our baseline model, we include a fixed intercept (or offset) in the full model which is the linear predictors for each subject and word pair that were estimated previously in the baseline model, i.e. $(\hat{\theta}_s + \hat{b}_w)$. The full model we estimate is therefore:

$$P(r_{ws} = 1 | s, w, \hat{\theta}, \hat{\mathbf{b}}, \mathbf{X}) = \text{logit}^{-1}((\hat{\theta}_s + \hat{\mathbf{b}}_w) + \beta \mathbf{x}_{ws}) \quad (1.1)$$

where $\mathbf{x}_{ws}$ is a vector of neural features (e.g. average activity in several ROIs) for subject s on word pair w .

Standard Subsequent Memory Model In addition to the full model, we also estimate a more typical subsequent memory model that simply predicts recall from the fMRI signals without controlling for the predictions from the baseline model. This model is as follows:

$$P(r_{ws} = 1 | s, w, \mathbf{X}) = \text{logit}^{-1}(\alpha + \beta \mathbf{x}_{ws}) \quad (1.2)$$

In order to prevent overfitting and maximize predictive accuracy for both models including fMRI signals, we fit the model using maximum likelihood with an L_2 penalty on

the β parameters, also known as ridge logistic regression (Hoerl and Kennard, 1970; Cessie and Houwelingen, 1992). This method has been used in several successful applications of MVPA techniques for studying the neural predictors of memory (e.g. Kuhl et al., 2012; Rissman et al., 2010; Weidemann and Kahana, 2019). Because the resulting predictor is a linear function of the features, it also allows for easy interpretation of successful classification, i.e. some subset of covariates used in the classifier had particular associations with the outcome. Our goal is to test whether features suggested in the literature do correspond to endogenous mechanisms and a linear classifier allows us to determine that a specific feature shown to correlate with memory in standard subsequent memory paradigms also does so in some subset of ROIs after controlling for memorability. In addition to interpretability, using a linear classifier has two advantages over nonlinear ones, like neural networks, that have had recent success in data-rich fields: it is more reasonable to expect to be able to estimate the parameters of a linear classifier than a nonlinear one in a small data set like ours that only contains on the order of 2,500 rows; there are also fewer hyperparameters to select from (such as structure of a neural network) and the optimization problem is convex which makes us more confident that success or lack thereof is not due to errors in model fitting.

To estimate the models, we use cyclical coordinate descent as implemented in the `glmnet` package (Friedman et al., 2010) in R. We estimate the predictive power of our models with leave-one-subject out cross-validation (Stone, 1974), using the area under the ROC curve as a performance metric. We choose the size of the penalty on the β parameters (λ) using nested 10-fold cross-validation. We choose a set of λ hyperparameters to try using the default algorithm in `glmnet`: first $\lambda_{\max}$ is chosen such that a model fit with that λ on all of the data will have all parameter estimates near 0, using a formula from Friedman et al. (2010).

Then 100 parameters are chosen on a log scale from $\lambda_{\max}$ to $\frac{\lambda_{\max}}{10000}$. The model is estimated for each λ and the λ that maximizes the log likelihood of the held-out fold is chosen for evaluation. All of our fMRI-derived features are standardized (by their mean and variance in the training fold) before being included in the model.

To evaluate the models' generalization performance, we conduct statistical tests on the held-out AUCs. For the standard models, we compute a one-sample t-test, comparing the model's performance to the AUC of a random guessing model (.5). AUC is a useful metric here because it is not affected by the base rates of remembering and forgetting. For the main test, we compare the full models to the behavioral model using a paired t-test. To give the strongest possible chance to the MRI features, for all models, we use a *one*-sided test of whether the AUC in the full model is greater. In leave-one(-subject)-out cross-validation, the distribution of AUC scores (or any other metric) across holdout sets will in general be correlated because the training set for the classifiers will be largely the same. The independence assumptions of the t-test are therefore violated (Dietterich, 1998). In order to remedy this, we use a permutation test to estimate the empirical null distribution of paired *t*-statistics when there is no relationship between the fMRI features and recall (Pereira and Botvinick, 2011; Pitman, 1937; Stelzer et al., 2013; Golland and Fischl, 2003). To do so, we shuffle the pairing between the fMRI features and the Lithuanian-English word pair within subject. This allows us to maintain the correlation between fMRI features, as well as the distribution of word-pair population memorabilities and subject recall abilities. Crucially, this also means that *Behavioral Baseline Model* does not change its predictions so the residual variance to be explained remains the same across permuted datasets. When shuffling the data, we also maintain the structure of the sequence of items such that the same five items

appear in each group of five serial positions of each study block. We created 500 permuted datasets, applied to same classification models and computed a paired t -statistic for the permutation test.

One final complication is the fact that each item was shown multiple times in our study. Therefore, for each item, we have several observations that need to be combined to make a prediction. For the activity and inter-subject pattern correlation features there was one observation for each item in each of the five study blocks. For the pattern similarity features, there were observations for all possible pairs of study blocks. In the item pattern similarity feature, there were ten possible combinations of study blocks since the same item appeared in both of the patterns being correlated. In the global pattern similarity there were 20 possible combinations since each item in one of the five study blocks could be compared to items in the other four study blocks. To deal with combining these features, we tried three different approaches:

Mean In our first approach, we simply took the mean of all of the observations as a single observation for that ROI and item. This approach has been used in previous studies with multiple presentations of the same item (e.g. [Xue et al., 2010](#)) and weighting a number of features equally can have good statistical properties over learning individual weights, especially in small datasets ([Dawes, 1979](#); [Schmidt, 1971](#)).

All In the second approach, we included all features on all repetitions, allowing the fitting and regularization to determine both which regions and which time points were the most relevant for prediction.

Individual blocks/block pairs In the final approach, we fit each study block or pair of

study blocks individually. This allows us for the possibility that the most relevant information comes only in earlier or later blocks. By not including as many potentially unrelated features as the second approach, the optimal amount of penalization is lower (Theobald, 1974). This means that the accuracy of the predictions will be higher. Within a feature and ROI set combination, we do a false discovery rate (FDR) adjustment to account for the 5, 10 or 20 tests that are done on each combination (Benjamini and Hochberg, 1995). Our tests are likely to be correlated within feature set across study blocks but this positive correlation is accounted for in the standard FDR procedure (Benjamini and Yekutieli, 2001). This makes these tests more comparable to the individual model tests using the other approaches.

As a comparison to the predictive models using the fMRI-derived features, we also test a model using the Judgements of Learning as a predictor of subsequent memory. For this model we simply use the raw numbers input by the subjects with no further preprocessing.

1.3 RESULTS

1.3.1 BEHAVIORAL RESULTS

There was a large amount of variability in both subject memory abilities and word memorability scores in our data. Taking the average within each word pair, the median word pair was remembered by 37% of subjects with a standard deviation of 15% across all 45 word pairs. Taking the average within subjects, the median subject remembered 33% of the word pairs with a standard deviation of 26% across all 57 subjects. The distributions of word pair memorabilities and subject memory abilities are shown in figure 1.2

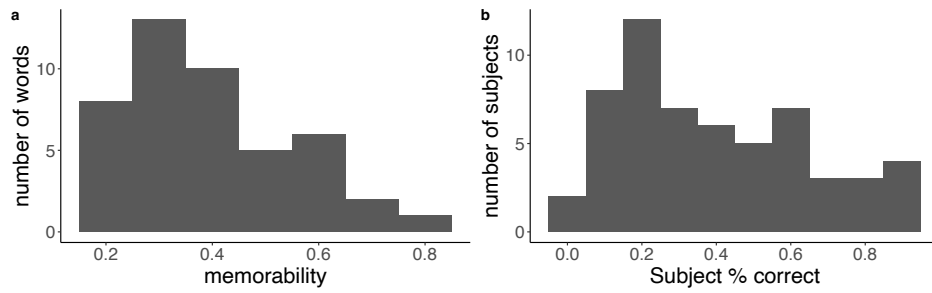


Figure 1.2: Panel a shows the distribution of memorabilities (percent of subjects remembering the pair in a cued recall test) across all 45 word pairs. Panel b shows the distribution of subject abilities over all 57 subjects

1.3.2 CLASSIFIER RESULTS

We first test whether our *Standard Subsequent Memory Model* can predict recall above chance using our proposed features and no other information. In Figure 1.3, we plot the mean and standard error of the AUC estimates in held-out subjects. We show results using both the mean feature across all repetitions as well as using all study block and study block pairs. Several of our feature/ROI combinations, when aggregated at the mean level, predict memory significantly above chance (.5) based on a permutation test. These include all models including all Schaefer et al. (2018) ROIs, all models using *Targeted* ROIs except global pattern similarity (GPS) and GPS within the hippocampal subparts ($M = .538-.624$, $t_{56} = 2.59 - 7.75$, $ps < .01$). Results are overall similar when including each time point separately except that the Schaefer et al. (2018) intersubject pattern correlation (ISPC) and the hippocampal subparts model are no longer significant while the GPS model involving the whole hippocampus now is ($M = .527-.6$, $t_{56} = 1.83 - 6.5$, $ps < .05$). Because we are testing so many models it may make sense to control the overall false discovery rate across all models tested. When we calculate the adjusted q values, the hippocampal models are no longer significant, along with the *Targeted* mean GPS model and the Schaefer et al. (2018)

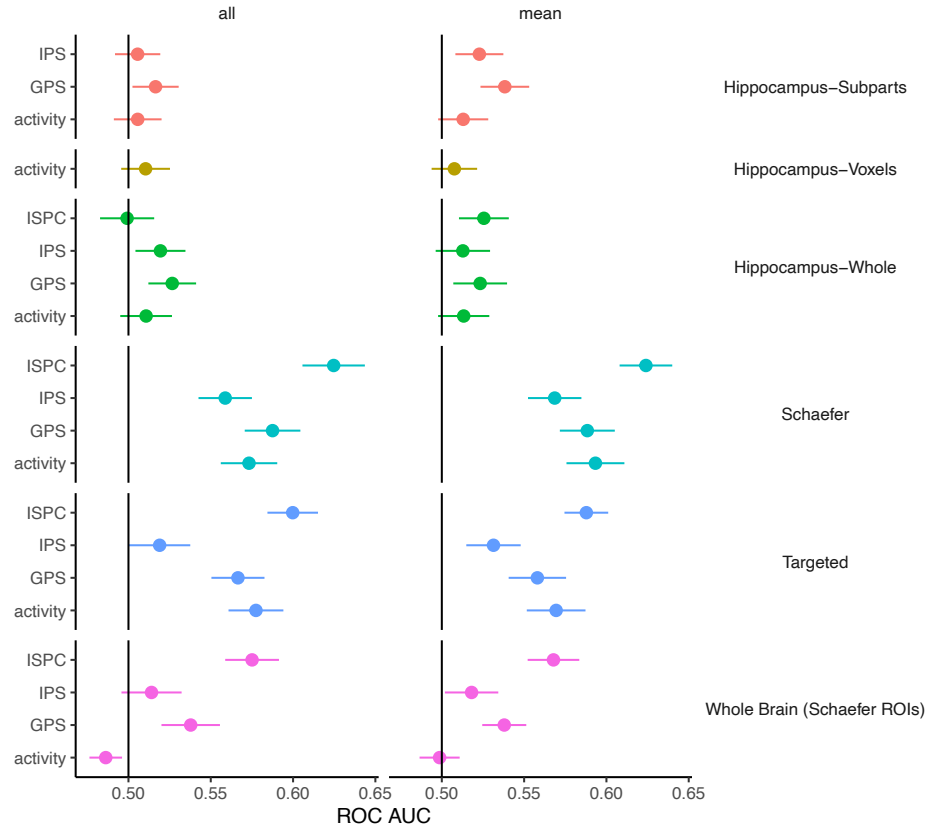


Figure 1.3: Classifier performance based on the *Standard Subsequent Memory Model*. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation

item pattern similarity (IPS) and ISPC models including all features.

Overall, this suggests that several features we tested are able to successfully predict memory across subjects and would be classified as subsequent memory effects in typical studies. In particular, when averaging over all study blocks and including all cortical ROIs, all of the features we derived from the literature could successfully predict memory in a heldout subject.

We now turn to the *Full Model* that controls for the memorability of individual items

using the *Baseline Behavioral Model*. Figure 1.4 shows the results of the two models including the mean feature across all study-blocks and including all features. Figure 1.5 plots the results of using only a single study-block or study block pair for each classifier. We compare each model’s held-out AUC performance to the baseline model using a paired t -test. None of our fMRI feature-based classifiers using the mean of the study-blocks or all study-blocks/study-block pairs perform significantly better than the *Baseline Behavioral Model* (all $p > .07$). In addition, after False Discovery Rate (FDR) adjustments (Benjamini and Hochberg, 1995), none of the classifiers using only a single feature predict significantly better than the *Baseline Behavioral Model* (all $q > .1$).^{||}

In contrast to the classifiers with fMRI-derived features, the classifier that included judgments of learning did predict significantly better than the baseline model, improving the AUC on average by .0128 ($t_{(56)} = 6.84, p < .002$).^{**}

1.4 GENERAL DISCUSSION

We found that several univariate and multivariate features of fMRI data proposed in the subsequent memory literature (Kim, 2011; Xue, 2018) could predict cued recall over a long delay in a verbal paired associates task. To our knowledge, this is the first demonstration of subsequent memory effects in such a task. We also provide the first demonstration that Intersubject Pattern Correlation (ISPC) is a predictor of memory in a task not using video

^{||}Two of the classifier models, one including all hippocampal voxels from the first study block and one including the hippocampal inter-subject pattern correlation have undefined comparisons because they had minimal variation related to recall and the cross-validated penalty that was chosen meant that these features did not contribute to the prediction. Therefore, the difference between the AUCs in the baseline model and the two fMRI models was 0 and the variance was undefined, leading to an undefined t -statistic.

^{**}This was the smallest possible p -value in our permutation test given that we ran 500 samples. The true p -value may be much smaller but we do not have the precision to report it.

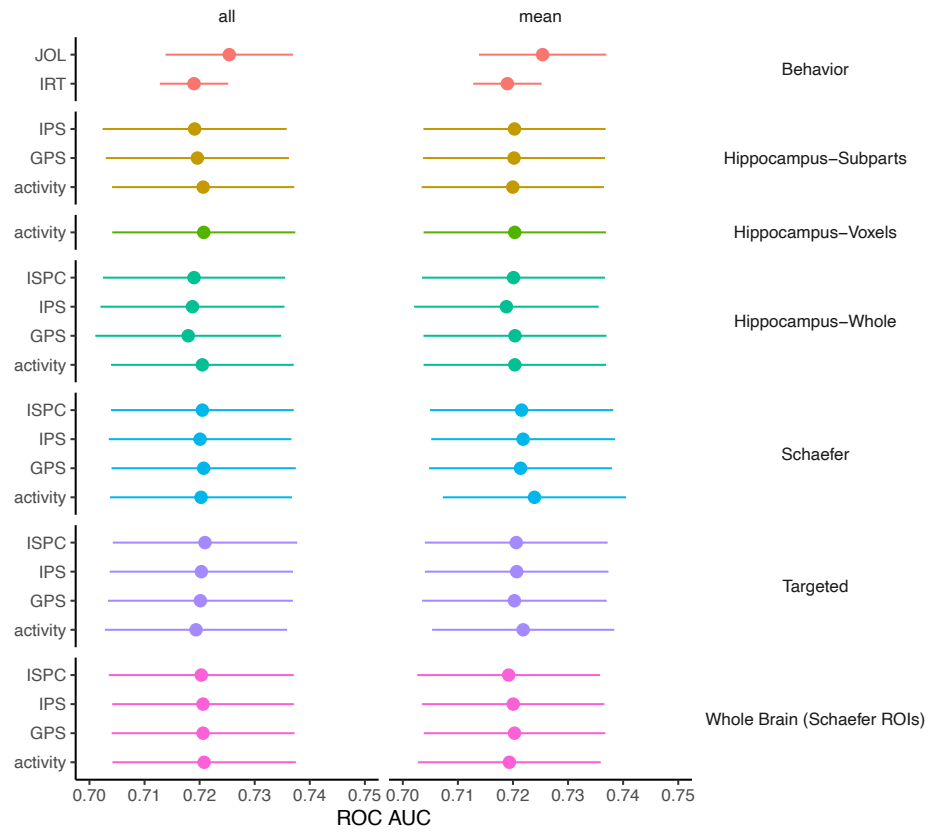


Figure 1.4: Classifier performance based on the *Full Model*. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.

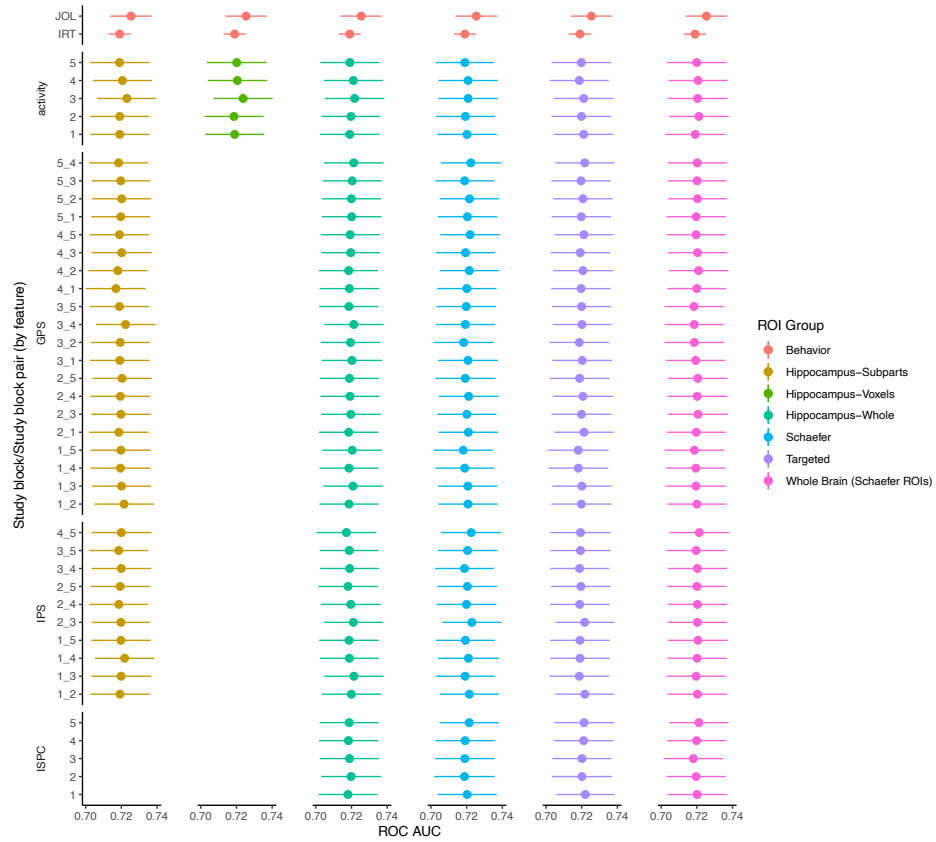


Figure 1.5: Full Model results by study block. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.

stimuli and that Global Pattern Similarity outside of hippocampus predicts memory in a recall task.

However, like many memory experiments, the items that we chose had a large amount of variation in their average memorability (Bainbridge, 2019). We therefore tested to what extent these effects could be explaining variation in recall that is not explained by population memorability. A common perspective on subsequent memory effects is that they index “the depth of encoding of the to-be-remembered stimuli” and “determine the efficacy of memory encoding” (Fukuda and Woodman, 2015). In contrast, our analyses, based on comparing predictive models using only behavior with models using behavior and fMRI data combined, showed that the subsequent memory effects we observed were highly correlated with population memorability and did not consistently improve predictions across subjects once the predictive model included the population memorability scores. This suggests that the interpretation of the source of subsequent memory effects may be more complicated than typically assumed. There has long been a recognition that “a specific experience may elicit the recruitment of the [processes indexed by subsequent memory effects] to a greater or lesser extent because of variable task demands, shifts in subjects’ strategies, characteristics of target items, or attentional modulations” (Wagner, 1998). At least in our data, these analyses suggest that objective characteristics of target items may drive a significant portion of the subsequent memory effect.

Of course, our analyses do not rule out the other sources of variability, particularly given that, in our experimental design, we did not explicitly manipulate task demands, nor impose a strategy on subjects. In addition, the nature of our task (long delays, associative memory, novel stimuli), may have prevented some processes from being involved that may

typically be measured by subsequent memory effects. However, the strength of the results using the *Standard Model* and the results of the *Full Model* including the judgments of learning place limits on these alternative explanations. In the following, we discuss several interpretations of these analyses and their implications for future studies of subsequent memory.

One explanation is that much of the variability in memory that could be predicted at encoding is explained by memorability already. For the fMRI features to predict when controlling for memorability, they would have to be reliably indexing endogenous processes that contradict the prediction from memorability. For example, a consistent failure of attention on certain trials for easy items that was evident in the neural signals would improve their predictive power. In addition, due to the long delay, there is the potential for a lot of variability in memory accessibility between encoding and retrieval due to events that happen in between, limiting the predictability of memory in our task. However, the performance of the model including Judgments of Learning shows that there is a signal (that is in fact consciously available to subjects) that consistently predicts subsequent memory even across items with the same population memorability. This model gives a lower bound on what is explainable at the time of encoding.

A second possible interpretation is that there is too much noise in the fMRI data to hope to find signals comparable to a judgment of learning in a data set our size. However, at 57 subjects, this dataset is approximately 2-3 times the size of most data sets in the subsequent memory literature, indicating that future work trying to isolate endogenous shifts in encoding processes may require large datasets and tightly-controlled designs.

A third possible interpretation is that there are indeed predictive signals that exist in

fMRI data but they are highly variable across people, perhaps due to a lack of precise voxel alignment or differently shaped brains causing variation in the signal-to-noise characteristics. If this were the case, it may be that our attempt to do across-subject prediction was doomed from the start. However, we point to the success of the *Standard Model* as evidence that these issues do not prevent any across-subject prediction in our data. It may be interesting to try to explain in future work why voxels that are correlated with memorability seem to be less sensitive to this type of variation. In addition, several of our features made efforts to circumvent this difficulty by a) aggregating data at higher levels (e.g. the whole brain analyses treating each ROI as a “voxel”), b) using hippocampal ROI definitions based on individual subjects’ anatomy and c) using multivariate measures like pattern similarity that are less sensitive to perfect alignment across subjects. However, none of these seem to have succeeded in improving classifier accuracy significantly.

A fourth possible interpretation is that the actual cognitive strategies used for learning (especially in an undirected, intentional encoding task like ours) are highly variable across people. For instance, it may be that high memorability word pairs are easier to sound out, resulting in greater activity in phonological regions on average. But if sounding out is only an effective strategy for some people activity in a phonological region may predict memory but not consistently beyond the memorability score. Across-subject prediction differs from the usual two-level within-subject paradigm that is typical of MVPA studies in that it is more sensitive to small magnitude, low variability effects but less sensitive to high magnitude, high variability effects (Wang et al., 2020). If the measurement of encoding processes is more like the latter scenario, this suggests (like above) that future studies of subsequent memory will require larger data sets that allow for understanding groups of subjects who

use similar encoding strategies.

Overall, these four explanations can be summed up by a low signal-to-noise ratio and/or high signal but high across-subject variability in both the fMRI BOLD measurement and the cognitive processes involved in encoding themselves. We argue that the existence of the JOLs as an orthogonal predictor to memorability and the existence of successful memory prediction from fMRI in the absence of memorability information suggest that these can't be the complete story.

One additional possibility is that endogenous processes indexed by subsequent memory signals might not be simply randomly fluctuating but may actually fluctuate strategically in a way that is related to memorability. For instance, a subject might decide whether or not to allocate attention to a particular item based on whether they think it is possible to learn the item. We might hope that there would be some variation in this perceived memorability that is not captured by the population memorability but could be captured by fMRI signals. For instance, two words in a paired associates task might not be similar according to most people but might have been more closely paired based on a person's individual experiences (Charest et al., 2014). However, it is possible that controlling for memorability removes too much of this variance for it to be detectable in our data.

1.5 SUMMARY

In the forty years since the first subsequent memory effect was reported, much has been learned about the neural signals associated with memory. A vast literature documents the various regions and multivariate signals, along with proposed cognitive computations being implemented and supporting evidence from animal and lesion studies. This paper con-

tributes to this literature by documenting, for the first time, several aspects of subsequent memory effects in a novel language-learning task. We show that several features proposed to be relevant for other types of memory tasks such as intersubject pattern correlation and global pattern similarity also apply here. However, the exact interpretation of these subsequent memory effects remains ambiguous and the original hope of identifying an item-level measure of depth-of-processing has still not been fully realized. Here we attempted to precisely control for stimulus and presentation-order effects that have been documented to have influence over the probability of recall in order to see whether we can extract signals from fMRI that correlate with moment-to-moment fluctuations in ability to encode. While we were not able to identify predictive signals, we suggest that future work leverage the techniques of this paper and [Weidemann and Kahana \(2019\)](#), novel classifier analyses and tight control of memorability, with larger datasets, different tasks and new features of the fMRI signal. In several ways, our own design could be improved, for instance by using items with more similar memorability scores (so that more variance can be explained by the neural signals), by including a movie-watching portion of the scanner task that would allow for the use of better alignment tools ([Haxby et al., 2011, 2020](#)) or by using a task such as recognition memory that would allow for more trials to be collected per participant. Despite these qualifications, we think that these results suggest that interpretations of classic subsequent memory results may not be as straightforward as commonly assumed and lay out key questions for cognitive neuroscientists of memory to address in the future.